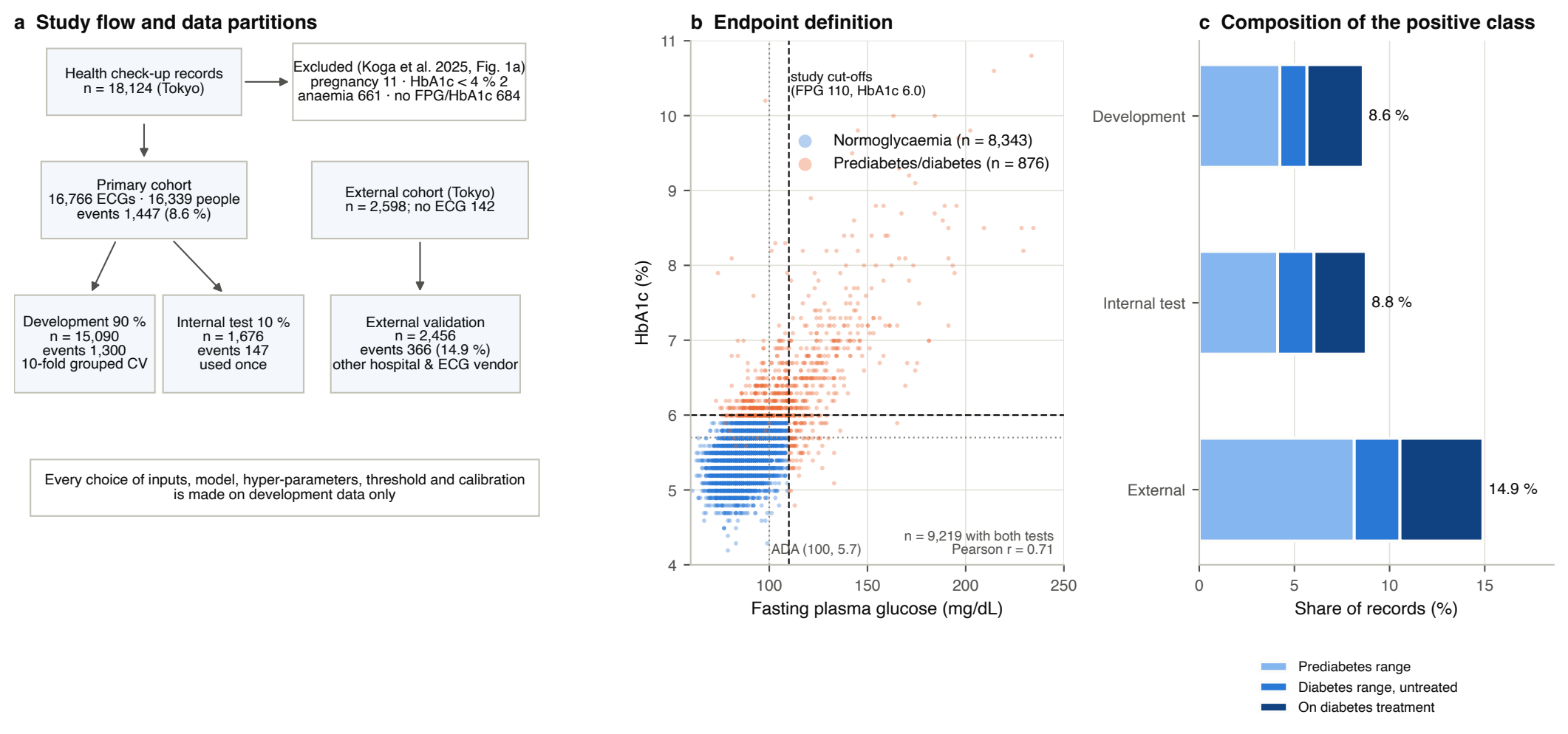

a Study flow and data partitions
Health check-up records
n = 18,124 (Tokyo)
Excluded (Koga et al. 2025, Fig. 1a)
pregnancy 11 · HbA1c < 4 % 2
anaemia 661 · no FPG/HbA1c 684
Primary cohort
16,766 ECGs · 16,339 people
events 1,447 (8.6 %)
External cohort (Tokyo)
n = 2,598; no ECG 142
Development 90 %
n = 15,090
events 1,300
10-fold grouped CV
Internal test 10 %
n = 1,676
events 147
used once
External validation
n = 2,456
events 366 (14.9 %)
other hospital & ECG vendor
Every choice of inputs, model, hyper-parameters, threshold and calibration is made on development data only
b Endpoint definition
study cut-offs
(FPG 110, HbA1c 6.0)
Normoglycaemia (n = 8,343)
Prediabetes/diabetes (n = 876)
ADA (100, 5.7)
n = 9,219 with both tests
Pearson r = 0.71
HbA1c (%)
Fasting plasma glucose (mg/dL)
c Composition of the positive class
Development
Internal test
External
8.6 %
8.8 %
14.9 %
Share of records (%)
Prediabetes range
Diabetes range, untreated
On diabetes treatment